30 Matrix Common elements in all rows of a given matrix

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// A Program to prints common element in all
// rows of matrix
#include <bits/stdc++.h>
using namespace std;
// Specify number of rows and columns
#define M 4
#define N 5
// prints common element in all rows of matrix
void printCommonElements(int mat[M][N]
   unordered_map<int, int> mp;
   // initialize 1st row elements with value 1
   for (int j = 0; j < N; j++)
      mp[(mat[0][j])] = 1;
   // traverse the matrix
   for (int i = 1; i < M; i++
      for (int j = 0; j < N; j++)
      {// If element is present in the map and
         // is not duplicated in current row.
         if (mp[(mat[i][j])] == i)
         {
           // we increment count of the element
           // in map by 1
            mp[(mat[i][j])] = i + 1;

            // If this is last row
            if (i==M-1 && mp[(mat[i][j])]==M)

// driver program to test above function
int main()
{
   int mat[M][N] =
   {
      {1, 2, 1, 4, 8},
      {3, 7, 8, 5, 1},
      {8, 7, 7, 3, 1},
      {8, 1, 2, 7, 9},
   };
   printCommonElements(mat);
   return 0}
```